

Course Code	: 2304CS302	Date	: 25-10-2024
Course Name	: Object Oriented Programming using Java	Duration	: 150 Minutes
		Total Marks	: 70

Instructions:

1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

- Q.1 (A)** Write short note on bytecode in java. **4**
- (B)** Describe type casting with example. **3**

OR

Discuss ternary operator using java.

- (C)** Explain all java features. **7**

OR

Explain operator precedence and associativity with suitable example.

- Q.2 (A)** What is array? Write a java program to store element in array and display that element. **4**
- (B)** Demonstrate switch-case statement with suitable example. **3**

OR

Differentiate entry controlled and exit controlled loop.

- (C)** Write a java program to print Fibonacci series up to given number. **7**

OR

Write a java program to check whether the given number is prime or not.

- Q.3 (A)** Compare POP and OOP. **4**
- (B)** Implements any three methods of Math class. **3**

OR

Differentiate String and StringBuffer class by implementing only one function of it.

- (C)** List and explain all the OOPs concepts. **7**

OR

Explain types of inheritance in java.

- Q.4** **(A)** Demonstrate the use of static keyword. **4**
 (B) Explain function overloading. **3**

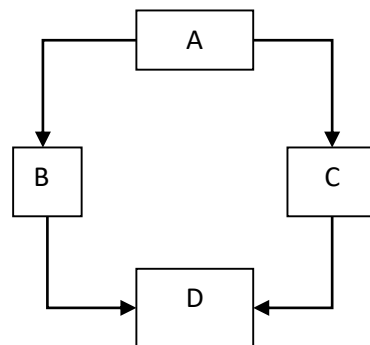
OR

Explain function overriding.

- (C)** Demonstrate all use of final and super keyword with example. **7**

OR

Compare the class and interface. Implement the following structure of inheritance using java.



- Q.5** **(A)** Write a java program to read data from one file and write it in to another file. **4**
 (B) Explain types of exception. **3**

OR

Explain the user defined exception (Customized Exception) using java.

- (C)** What is exception? what is the need of exception handling mechanism? **7**
 Implement exception handling mechanism with suitable example.

OR

Define a class A in package apack. In class A, three variables are defined of access modifiers protected, private and public. Define class B in package bpack which extends A and write display method which accesses variables of class A. Define class C in package cpack which has one method display () in that create one object of class A and display its variables. Define class ProtectedDemo in package dpack in which write main () method. Create objects of class B and C and class display method for both these objects.
